

Global Energy Transition & Climate Impact Dashboard

An Interactive Data Visualization Dashboard for Global Energy Transition and Climate Intelligence

INTRODUCTION

The transition from fossil fuels to cleaner and renewable energy sources has become one of the most significant global priorities in addressing climate change and ensuring sustainable development. However, understanding the relationship between energy consumption, renewable adoption, carbon emissions, and climate impact requires analysing large volumes of complex data collected from multiple international sources. This project develops an interactive Tableau dashboard that transforms global energy and greenhouse gas datasets into meaningful visual insights, enabling users to explore worldwide energy transition patterns, carbon emissions, and historical climate impact through intuitive visualizations.

PROBLEM STATEMENT

Global energy and climate datasets are extensive, multidimensional, and often difficult for students, researchers, policymakers, and the general public to interpret. Although valuable information is available through international organizations, extracting meaningful insights requires significant data processing and analysis. An interactive dashboard is needed to integrate these datasets and present them through intuitive visualizations, enabling users to compare countries, identify sustainability trends, and better understand the relationship between energy transition and climate change.

OBJECTIVES

- Develop an interactive Tableau dashboard integrating global energy and greenhouse gas emission datasets.
- Visualize renewable energy adoption, primary energy consumption, and global energy mix across countries.
- Analyse carbon emissions, carbon intensity, and historical contributions to global warming.
- Compare countries using interactive maps, KPI cards, trend analysis, scatter plots, and ranking charts.
- Enable users to explore sustainability patterns through dynamic filters and dashboard navigation.
- Support data-driven understanding of the global energy transition and climate change.

DATASET DESCRIPTION

Source Dataset: *Our World in Data (OWID) – Energy & CO₂ Database (2000–2024).*

The integrated dataset combines internationally recognized energy and greenhouse gas indicators, including:.

- Primary Energy Consumption
- Renewable Energy Share
- Fossil Fuel Share
- Low Carbon Energy Share

- CO₂ Emissions
- CO₂ Emissions per Capita
- Carbon Intensity of Electricity
- Greenhouse Gas Emissions
- Historical Contribution to Global Warming
- Population
- Country and Year

The dashboard integrates multiple OWID datasets using common identifiers (Country ISO Code and Year) to provide a comprehensive view of global energy transition and climate performance.

PROPOSED DASHBOARD FEATURES

Dashboard 1 – Global Energy Transition Dashboard

- **KPI Cards** – Total Countries Analysed, Total Primary Energy Consumption, Renewable Energy Share, Total CO₂ Emissions.
- **Interactive Map** – Global renewable energy adoption by country.
- **Bar Charts** – Top countries by primary energy consumption.
- **Lollipop Chart** – Top countries by renewable energy share.
- **Energy Mix Analysis** – Comparison of global energy sources.
- **Trend Analysis** – Global transition from fossil fuels to renewable and low-carbon energy.
- **Interactive Filters** – Country and Year selection.

Dashboard 2 – Climate Impact & Carbon Intelligence

- **KPI Cards** – Total CO₂ Emissions, Average CO₂ per Capita, Total Greenhouse Gas Emissions, Average Carbon Intensity.
- **Scatter Plot** – Renewable energy adoption versus CO₂ emissions per capita.
- **Bar Charts** – Top CO₂ emitting countries.
- **Carbon Intensity Analysis** – Countries with the highest carbon intensity of electricity.
- **Historical Climate Impact** – Countries contributing most to global warming.
- **Dashboard Navigation** – Interactive navigation between both dashboards.
- **Key Insights Panel** – Executive summary of analytical findings.

EXPECTED OUTCOMES

- Simplifies interpretation of complex global energy and climate datasets through interactive visualization.
- Enables comparative analysis of countries based on renewable energy adoption, emissions, and sustainability indicators.
- Helps identify global energy transition trends and regional climate challenges.
- Supports academic research, classroom learning, sustainability awareness, and evidence-based decision making.
- Demonstrates practical application of Business Intelligence, data integration, and interactive dashboard development using Tableau.

CONCLUSION

The **Global Energy Transition & Climate Impact Dashboard** transforms complex international energy and greenhouse gas datasets into an interactive decision-support platform that enables users to explore renewable energy adoption, carbon emissions, and historical climate impact across countries. By integrating multiple datasets and presenting them through intuitive visualizations, the dashboard promotes data-driven understanding of global sustainability challenges while demonstrating the practical application of Business Intelligence, data visualization, and analytical storytelling.

Link:

https://public.tableau.com/app/profile/shreya.mishra1905/viz/Global_Energy_Transition_Dashboard/GlobalEnergyTransitionDashboard?publish=yes